

BIOINFORMATICS

WEEK 3 – DAY 15

GENE ONTOLOGY (GO) RESOURCE

- world's largest, standardized database providing a computationally tractable vocabulary to describe the functional characteristics of genes and gene products across all species.
- Managed by the Gene Ontology Consortium.
- It enables bioinformaticians to interpret large-scale genomics data by categorizing biological data into three structured, independent domains.

Gene Ontology is a structured system that describes:

1. What a gene does.
2. Where it acts.
3. Which biological processes it participates in.

THREE GO CATEGORIES

Category	Focus Question	Definition	Scale of Activity	Examples (e.g., Human TP53)
Molecular Function (MF)	<i>What chemical or biochemical action does it perform?</i>	The elemental tasks or chemical reactions a gene product executes at the molecular level, independent of time and space.	Single-molecule level (Individual biochemical interactions)	• DNA binding • zinc ion binding • protein domain specific binding
Biological Process (BP)	<i>What broader pathway or objective</i>	A coordinated sequence of molecular events or operations leading	Network/System level (Requires multiple gene products)	• regulation of cell cycle

	<i>does it support?</i>	to an integrated cellular or organismal program.		• signal transduction in response to DNA damage • intrinsic apoptotic signaling pathway
Cellular Component (CC)	<i>Where inside or outside the cell does it operate?</i>	The exact macromolecular structures, complexes, or intracellular compartments where a gene product resides and executes its function.	Anatomical/Structural level (Spatial location within the cell)	• nucleus • nucleoplasm • transcription factor complex

WHY GO MATTERS?

- **Standardizes Text:** Replaces messy synonyms with a single, universal language for all organisms.
- **Maps Species:** Connects model organism data (e.g., mice) directly to human gene functions.
- **Simplifies Big Data:** Groups massive RNA-Seq gene lists into clear, readable biological pathways.
- **Aids Machine Learning:** Converts descriptions into structured, computer-ready graph data and numerical IDs.
- **Tracks Evidence:** Attaches clear codes proving if data comes from wet labs or software.